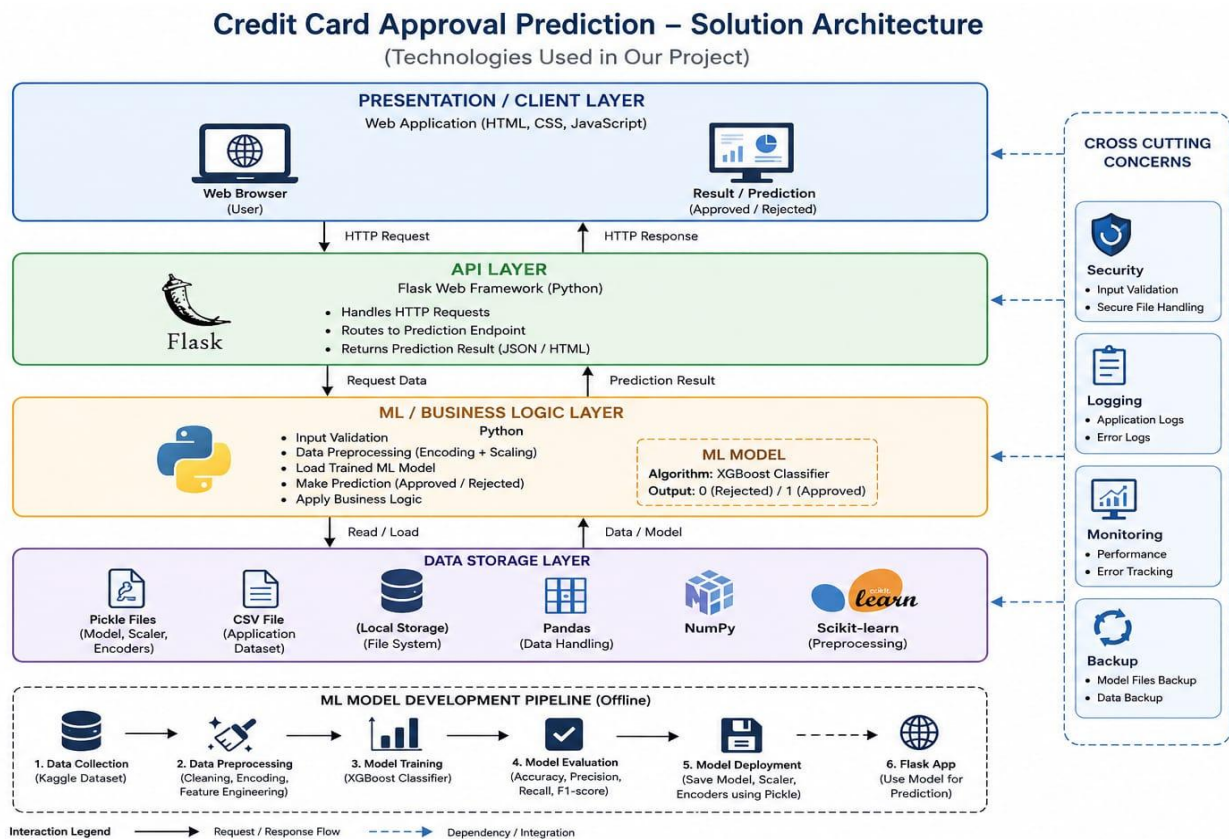


Solution Architecture

Date	29 JUNE 2026
Team ID	XXXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides the user interface where applicants enter their details and view the credit card approval prediction result.	HTML, CSS, JavaScript, Bootstrap
API Gateway	Receives user requests from the web interface, routes them to the Flask backend, validates inputs, invokes the ML model, and returns the prediction result.	<i>Flask (Python), HTTP</i>
Core Logic Service	Performs data preprocessing (label encoding and scaling), loads the trained XGBoost model, processes user inputs, generates the prediction, and returns the approval/rejection result.	Python, Flask, Pandas, NumPy, Scikit-learn, XGBoost, Pickle
Database / Storage	<i>Stores the trained machine learning model, scaler, label encoders, and the dataset used during model development.</i>	CSV Dataset, Pickle (.pkl) files, Local File System